

# A Modular Analytic Framework for the Riemann Hypothesis

(T1–T4)

Proof-only, journal-ready, arXiv-compatible

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## Abstract

Proof framework (T1–T4). This arXiv v1 tarball includes full proofs for T2, T3, and the appendix.

## 1 T2 — Weighted frame bound (full)

### 1.1 Weighted frame bound (T2): full proof

#### 1.1.1 Statement

Let  $\phi \in \mathcal{PW}$  be even,  $\text{supp } \hat{\phi} \subset [-A, A]$ ,  $\hat{\phi}(0) = 1$ .  
Let  $\Xi_A = \{\xi_{p,k} = k \log p \leq A\}$  and weights  $w_{p,k} = \frac{\log p}{p^{k/2}}$ .

Define  $G_{\alpha\beta}^{(w)} = \sqrt{w_\alpha} \hat{\phi}(\xi_\alpha - \xi_\beta) \sqrt{w_\beta}$ .

**Theorem 1.1.** *There exist  $A_0 \geq 1$  and  $\Theta_w(A) \in (0, 1)$  for  $A \geq A_0$  such that  $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A) =: C_{\text{fr}}^{(w)}(A) > 0$ .*

#### 1.1.2 Schur/Gershgorin

Let  $R = G^{(w)} - I$ . Schur's test gives

$$\|R\|_{2 \rightarrow 2} \leq \left( \sup_{\alpha} \sum_{\beta} |R_{\alpha\beta}| \right)^{1/2} \left( \sup_{\beta} \sum_{\alpha} |R_{\alpha\beta}| \right)^{1/2}.$$

Since  $R$  is symmetric,

$$\|R\|_{2 \rightarrow 2} \leq \Theta_w(A) := \sup_{\alpha} \sum_{\beta \neq \alpha} \sqrt{w_\alpha w_\beta} |\hat{\phi}(\xi_\alpha - \xi_\beta)|.$$

Hence  $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A)$ .

#### 1.1.3 Locality and summability

Fix  $\alpha = (p, k)$ . Band-limitation implies  $\hat{\phi}(\xi_\alpha - \xi_\beta) = 0$  unless  $|k \log p - \ell \log q| \leq A$ . Split  $\sum_{\beta \neq \alpha} = \sum_{q=p} + \sum_{q \neq p}$ .

**Same prime  $q = p$ .** Then  $|\ell - k| \leq \lfloor A / \log p \rfloor$  and  $\sqrt{w_{p^k} w_{p^\ell}} = (\log p) / p^{(k+\ell)/4} \leq (\log p) / p^{k/2} \cdot p^{-(\ell-k)/4}$ , whence

$$\sum_{\substack{\ell \neq k \\ |k-\ell| \leq A/\log p}} \sqrt{w_{p^k} w_{p^\ell}} |\hat{\phi}(\xi_{p,k} - \xi_{p,\ell})| \leq \frac{C_1}{p^{k/2-1/4}}.$$

**Different primes  $q \neq p$  (refined).** Locality forces  $q^\ell \in [p^k e^{-A}, p^k e^A]$ , so  $q \leq p^k e^A$  and the number of admissible primes is  $\ll p^k e^A / \log(p^k e^A)$  (PNT). For each  $q$ , admissible  $\ell$  lie in an interval of length  $\ll A / \log q$ . Using  $\sqrt{w_{p^k} w_{q^\ell}} \leq \sqrt{\log p \log q} p^{-k/4} q^{-\ell/4}$  and  $\sum_{q \leq X} \frac{\sqrt{\log q}}{q^{1/4}} = O(X^{3/4})$ ,

$$\sum_{q \neq p} \sum_{\ell \text{ admissible}} \sqrt{w_{p^k} w_{q^\ell}} |\hat{\phi}(\xi_{p,k} - \xi_{q,\ell})| \leq \frac{C_2 A}{p^{k/4}} (p^k e^A)^{3/8} = C_3 A p^{k/8} e^{3A/8}.$$

Weighted by  $w_\alpha = (\log p) / p^{k/2}$  yields  $\ll A e^{3A/8} p^{-3k/8}$ , geometrically small in  $k$ .

**Conclusion.** Thus

$$\sum_{\beta \neq \alpha} \sqrt{w_\alpha w_\beta} |\hat{\phi}(\xi_\alpha - \xi_\beta)| \leq \frac{C_1}{p^{k/2-1/4}} + \frac{C_3 A e^{3A/8}}{p^{3k/8}},$$

so  $\Theta_w(A) < 1$  for  $A \geq A_0$  and  $\lambda_{\min}(G^{(w)}) \geq 1 - \Theta_w(A) > 0$ .

## 2 T3 — Wave-packet localization (full)

### 2.1 Wave-packet localization (T3): full proof

Assume Theorem 1.1 and the explicit formula (Appendix). Let  $\gamma_0 \in (0, A)$  and suppose a nontrivial zero  $\rho_0 = \beta_0 + i\gamma_0$  with  $\beta_0 \neq \frac{1}{2}$ .

#### 2.1.1 Wave-packet construction

Pick  $b \in C_c^\infty([-1, 1])$ , even,  $\int b = 1$ . For  $T \geq 1$  set

$$\hat{\phi}_T(\gamma) = \frac{1}{2} \left( b(\sqrt{T}(\gamma - \gamma_0)) + b(\sqrt{T}(\gamma + \gamma_0)) \right).$$

Then  $\hat{\phi}_T$  concentrates near  $\pm\gamma_0$  with width  $\sim T^{-1/2}$ ; let  $\phi_T = \mathcal{F}^{-1} \hat{\phi}_T$ , with  $\|\phi_T\|_2 = 1$ .

#### 2.1.2 Frame capture and discrete-to-continuous limit

By T2,  $\sum_{\xi \in \Xi_A} w_{p,k} |\hat{\phi}_T(\xi)|^2 \geq C_{\text{fr}}^{(w)}(A)$ . Moreover:

**Lemma 2.1** (Frame-to-integral).

$$\sum_{\xi \in \Xi_A} w_{p,k} |\hat{\phi}_T(\xi)|^2 \longrightarrow \int_{-A}^A |\hat{\phi}_T(\gamma)|^2 d\gamma = 1 \quad (T \rightarrow \infty).$$

#### 2.1.3 Archimedean term $\rightarrow 0$ (explicit DCT)

Write  $A(\phi) = \int K(t) \hat{\phi}(t) dt$  with  $K(t) = 2 \log |t| + O(|t|^{-2})$ . Since  $\text{supp}(\hat{\phi}_T) \subset \{|t - \gamma_0| \leq c/\sqrt{T}\}$  and  $\|\hat{\phi}_T\|_1 = O(1)$ , choose

$$h(t) = C (1 + t^2)^{-1} (1 + \log^2(2 + |t|)) \mathbf{1}_{|t - \gamma_0| \leq 2}(t) \in L^1,$$

and note  $|K(t) \hat{\phi}_T(t)| \leq h(t)$  for  $T$  large; as  $\hat{\phi}_T(t) \rightarrow 0$  a.e., DCT gives  $A(\phi_T) \rightarrow 0$ .

#### 2.1.4 Prime–power side $\rightarrow 0$ (no square)

The explicit formula involves  $\sum_{p,k} (\log p)/p^{k/2} \phi_T(\xi_{p,k})$  (not squared). By Fourier inversion and repeated integration by parts, for any  $M \in \mathbb{N}$  there exists  $C_M$  with

$$|\phi_T(x)| \leq C_M \frac{T^{(M-1)/2}}{(1+|x|)^M}.$$

For fixed  $A$  there are finitely many  $\xi_{p,k} \in [0, A]$ , hence

$$\sum_{p,k} \frac{\log p}{p^{k/2}} |\phi_T(\xi_{p,k})| \leq \max_{\xi \in \Xi_A} |\phi_T(\xi)| \sum_{p,k: \xi_{p,k} \leq A} \frac{\log p}{p^{k/2}} \xrightarrow{T \rightarrow \infty} 0,$$

since  $\max_{\xi \in \Xi_A} |\phi_T(\xi)| \rightarrow 0$  (width  $T^{-1/2}$ , amplitude  $\rightarrow 0$  pointwise).

#### 2.1.5 Contradiction

The explicit formula yields

$$\sum_{\rho} \hat{\phi}_T(\gamma_{\rho}) = A(\phi_T) + \sum_{p,k} \frac{\log p}{p^{k/2}} \phi_T(\xi_{p,k}).$$

Letting  $T \rightarrow \infty$ , LHS  $\rightarrow 1$ , RHS  $\rightarrow 0$ ; contradiction.

### A Guinand–Weil explicit formula

For  $\phi \in \mathcal{PW}$  even, smooth, band-limited, contour-shift and residues for the completed  $\xi$ -function yield

$$\sum_{\rho} \hat{\phi}(\gamma_{\rho}) = \int K(t) \hat{\phi}(t) \frac{dt}{2\pi} + \sum_{p,k \geq 1} \frac{\log p}{p^{k/2}} \phi(k \log p),$$

with  $K(t) = \psi(\frac{1}{4} + \frac{it}{2}) + \psi(\frac{1}{4} - \frac{it}{2}) - 2\psi(\frac{1}{4})$  and  $\psi = \Gamma'/\Gamma$ .

### B Archimedean control (IBP)

Using  $\psi(\frac{1}{4} \pm \frac{it}{2}) = \log |t| + O(|t|^{-2})$  uniformly and a  $\cos^2$  taper with plateau+transition, two integrations by parts on the boundary layer give

$$|A(\phi)| \leq \frac{C}{A^2} \|\hat{\phi}\|_1.$$

### C Absorption lemma

If  $a \leq \eta a + b$  with  $\eta \in (0, 1)$ , then  $a \leq b/(1 - \eta)$ .

### References

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